KOGANTI SASANK

Senior Data Scientist and Machine Learning Engineer

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PROFESSIONAL SUMMARY

Senior Data Scientist with 4+ years of experience delivering predictive modelling, mathematical optimization and production ML systems for refinery, petrochemical and manufacturing operations. Built XGBoost models on million-row datasets, MILP scheduling solutions, real-time optimizers and reusable MLOps frameworks. Combines an M.Sc. in Mathematics with hands-on Python, SQL, model validation and production engineering.

CORE SKILLS

Machine learning: XGBoost, scikit-learn, supervised learning, predictive modelling, feature engineering, model validation, AutoML
Optimization: MILP, linear programming, differential evolution, JADE, constrained optimization, scheduling, Gurobi, CPLEX, PuLP
Data and programming: Python, SQL, Pandas, NumPy, PostgreSQL, MariaDB, Elasticsearch, large-scale data processing
Engineering and MLOps: FastAPI, Flask, Docker, Kubernetes, Jenkins CI/CD, MLflow, Kubeflow, pytest, Bandit, model quality gates, drift checks

PROFESSIONAL EXPERIENCE

Senior Data Scientist | TCG Digital, Remote, India | Jul 2025 - Present

- Lead productization of a real-time refinery optimizer using JADE-based differential evolution to maximize yield and operating margin under plant constraints; delivered a client-reported 1.5%-1.8% yield uplift and approximately INR 3,000-4,000 per ton-hour margin improvement.
- Designed the roadmap for ML and LLM-assisted initialization of optimization vectors from historical operating data to reduce convergence time and improve repeatability across deployments.
- Architected a grade-market and production-sequencing solution that combined market allocation with constrained scheduling, generating approximately INR 2 crore in estimated business benefit.
- Lead architecture, project planning and production delivery for data-science solutions, manage two team members, mentor 5-6 colleagues and coordinate with engineering and DevSecOps teams.
- Standardized automated testing, model quality gates and scalable API deployment practices across the data-science competency.

Data Scientist Consultant | TCG Digital, Remote, India | Jul 2022 - Jun 2025

- Developed an end-to-end real-time optimizer for a naphtha-cracker unit using a reusable JADE framework and multiprocessing, reducing optimization runtime by 90%.
- Built MILP-based grade scheduling and market-assignment optimizers for petrochemical and textile operations by translating production constraints, contracts and line capabilities into executable schedules.
- Delivered XGBoost models for drilling-fluid rheology and thickening-time prediction across 1M+ records and 50+ locations; evaluated global, local and grouped-location strategies and completed delivery in under two months.
- Created a reusable AutoML framework for feature engineering, model selection, validation and reporting, with optional TPOT, PyCaret and AutoGluon integration.
- Built a reusable CI/CD framework for ML services with linting, Bandit scans, pytest, golden-dataset regression checks and automated FastAPI and Flask deployment through Jenkins.

ADDITIONAL EXPERIENCE

Production AI and GenAI Systems

- Built production agentic RAG, patent intelligence and private LLM systems using LangGraph, Elasticsearch, Gemma, Llama 3.1, Ollama and vLLM.

LEADERSHIP AND RECOGNITION

Co-leads the Optimization and DevOps guilds and has created reusable components that let delivery teams define objectives and constraints without rebuilding the optimization framework.

EDUCATION

M.Sc. Mathematics | BITS Pilani | 2018 - 2022